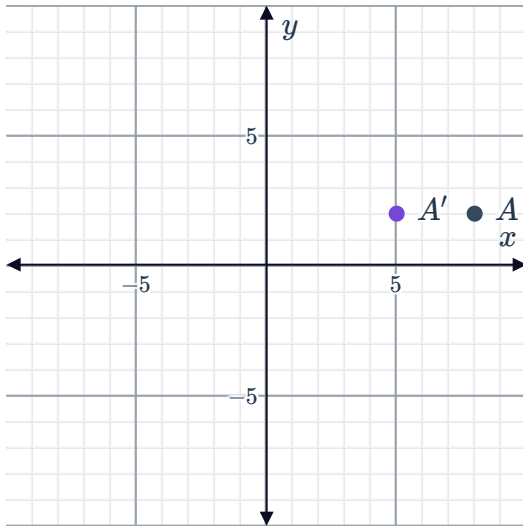


Compositions of transformations

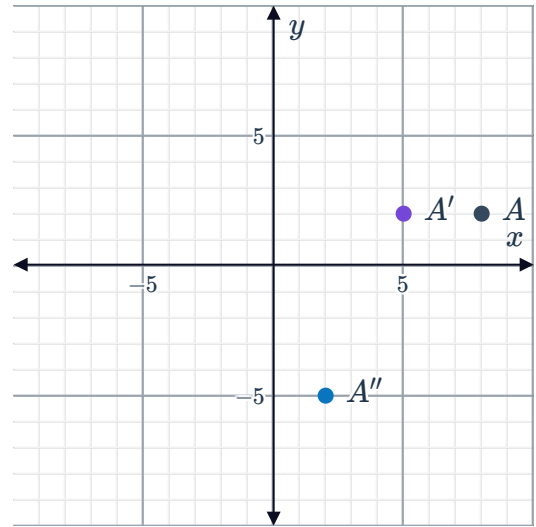


1

a

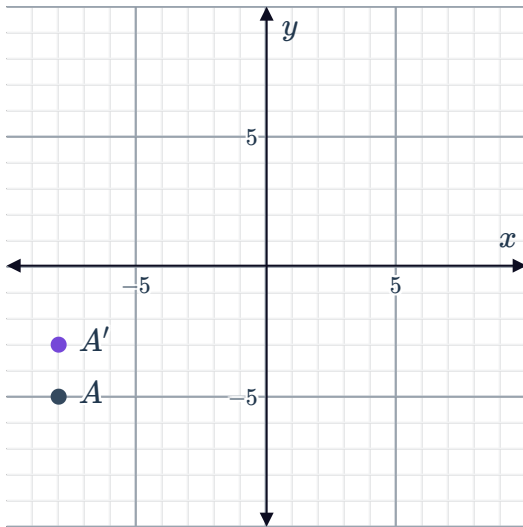


b

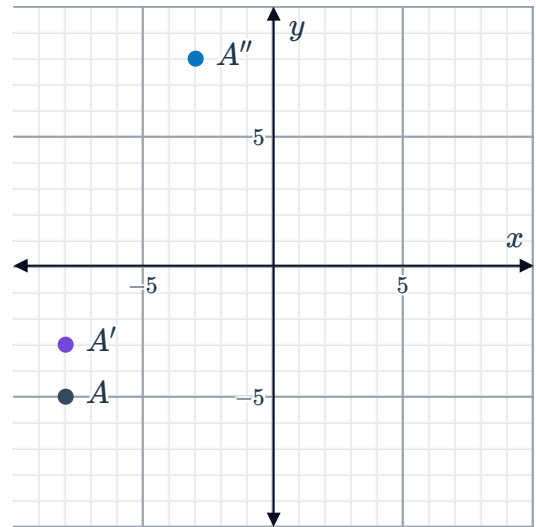


2

a



b

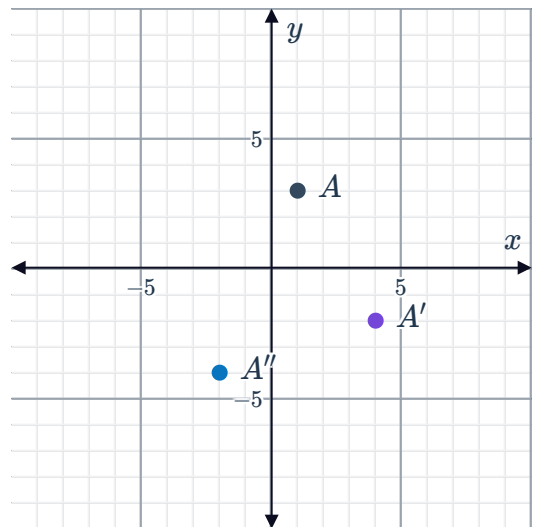
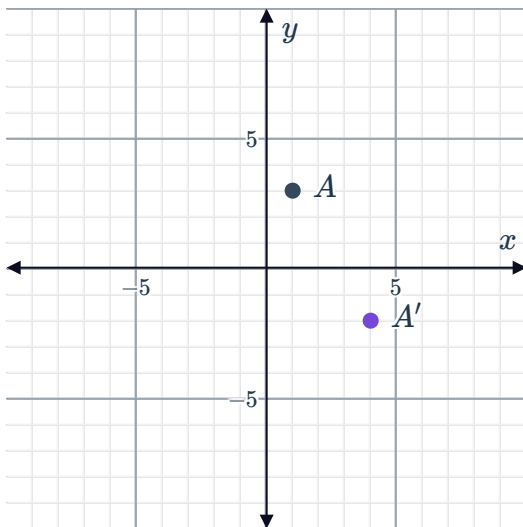


3

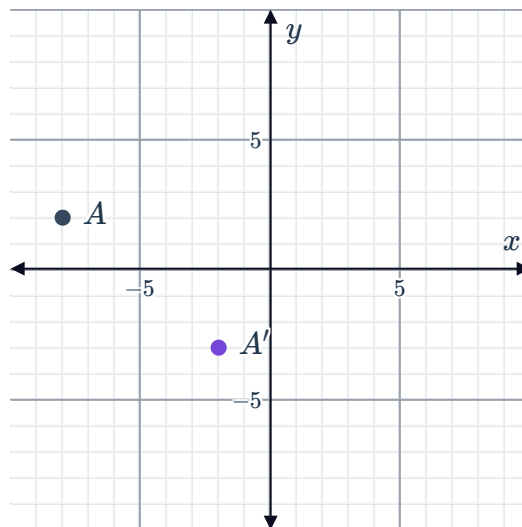
a



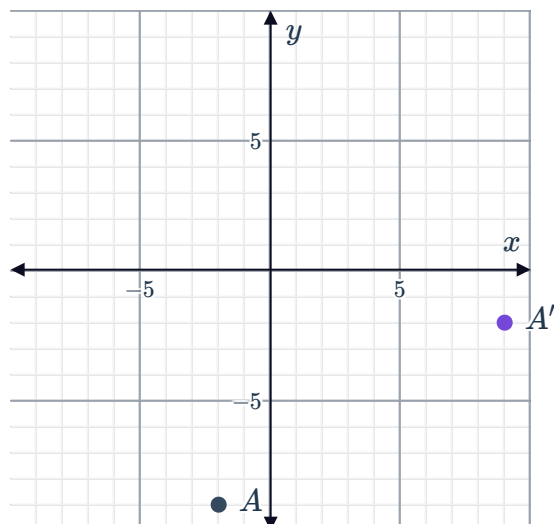
b



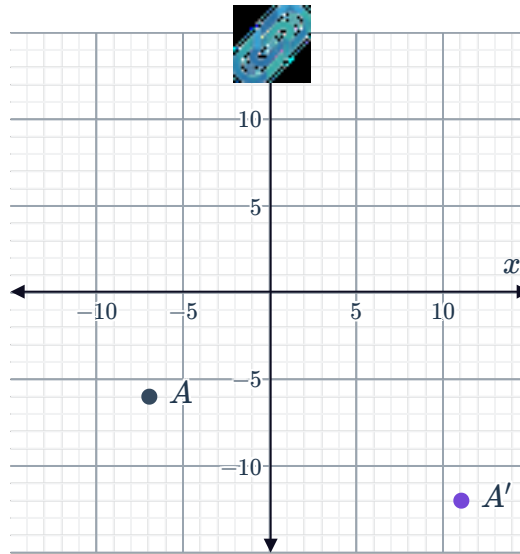
4



5



6



7

a $B(-4, -4)$

b $y = 0$

8

a $(-a, -b)$

b True

9

a $B(7, 9)$

b 16 units to the left

10

a $(-3, 9)$

b $(-4, 9)$

c $(3, -4)$

d $(-10, 3)$

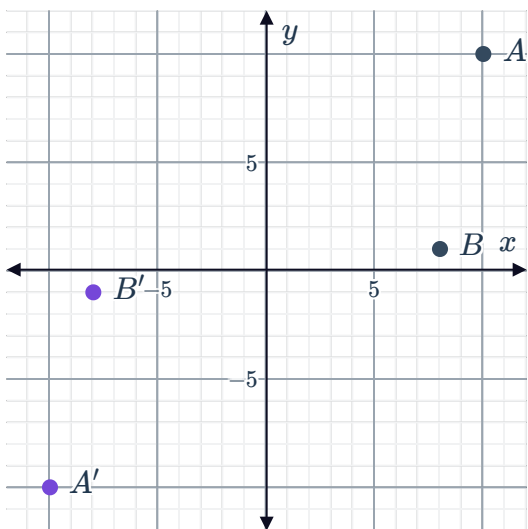
11

a $A'(-6, -1)$

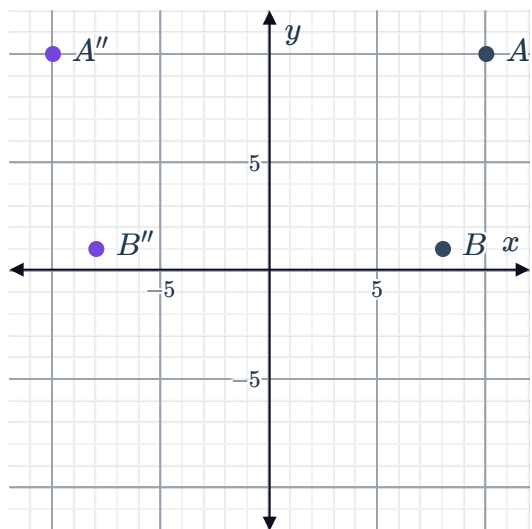
b
i Yes ii No iii No iv No

12

a



b

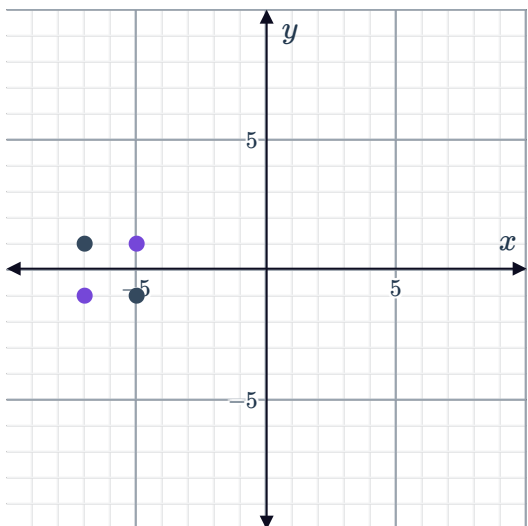


c

- i No ii Yes iii No iv No

13

a



b

- i No ii No iii No iv Yes

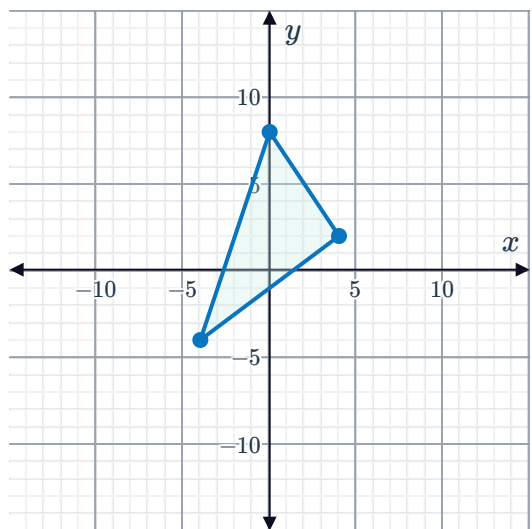
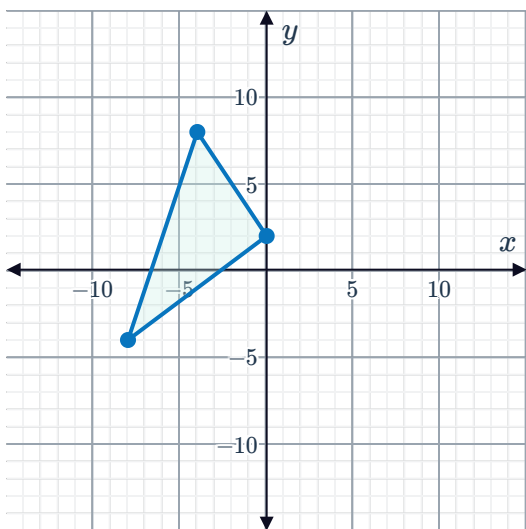
14

- a $A'(-10, 8)$
 $B'(-7, 7)$

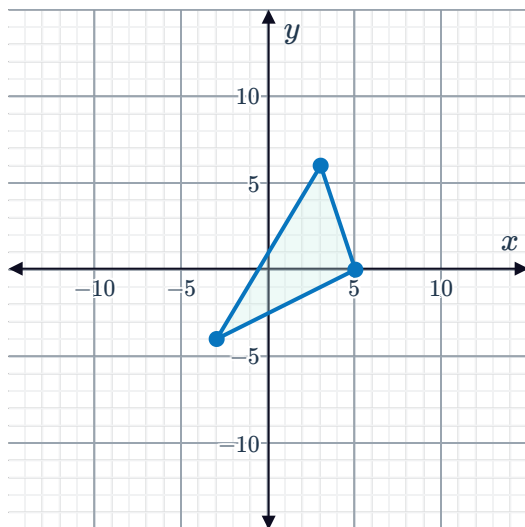
b

- i No ii Yes iii No iv No

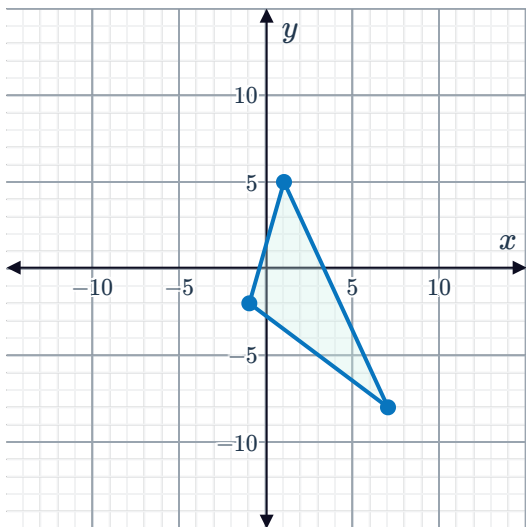
15
a



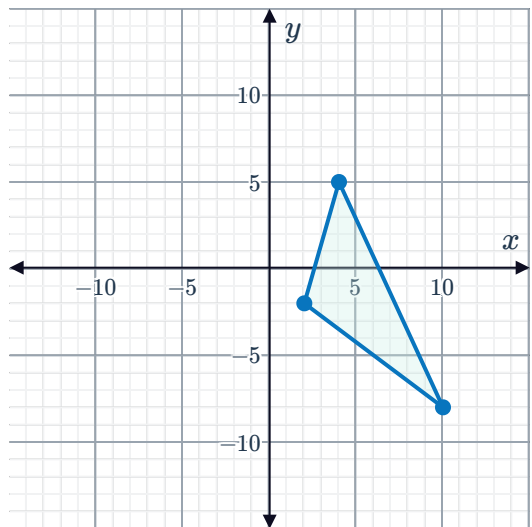
16



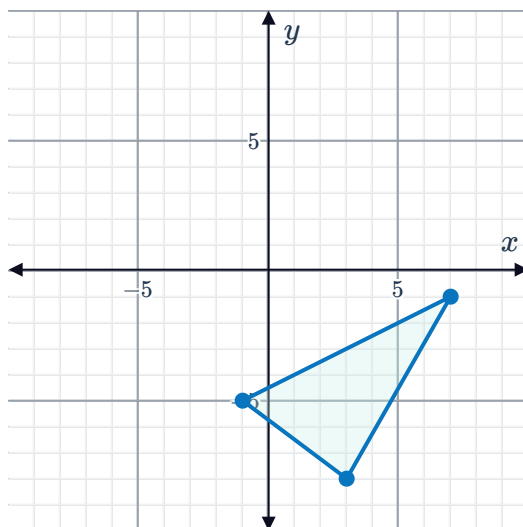
17
a



b

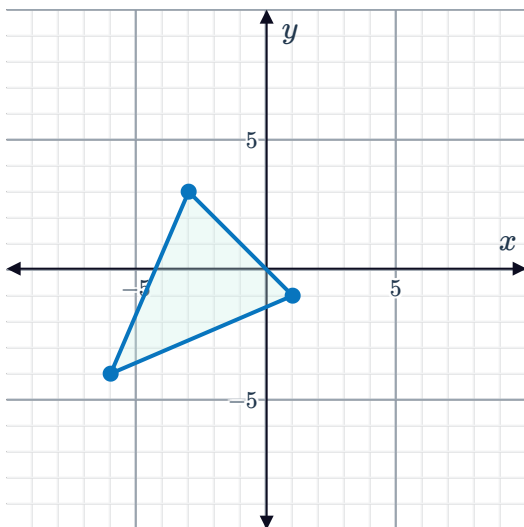


18

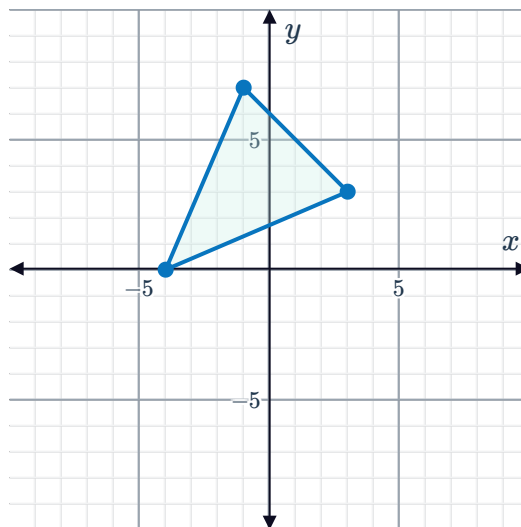


19

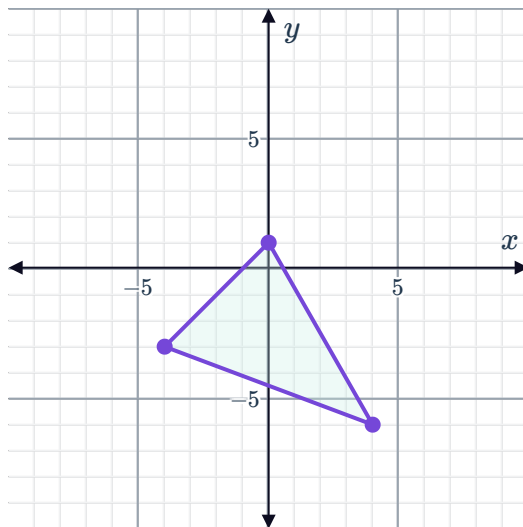
a



b

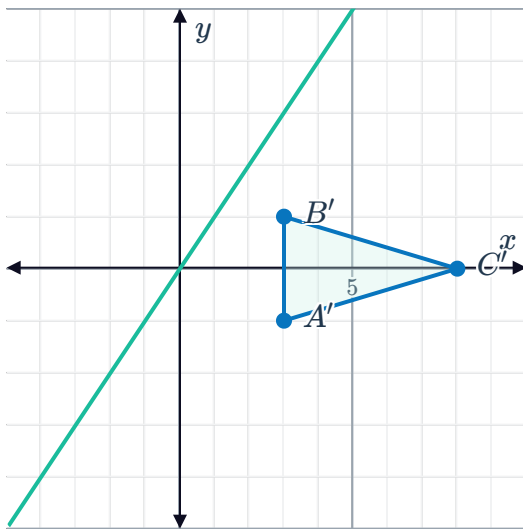


20



21

a



Possible answers:

A 90° rotation counterclockwise about the origin.A 270° rotation clockwise about the origin.

22 $A'(-3, -5)$

$B'(-1, 8)$

$C'(14, 10)$

23 $A'(-10, -7)$

$B'(5, 2)$

$C'(5, -1)$

24 $A'(1, -9)$

$B'(1, 3)$

$C'(13, 2)$

25 $A'(-8, -1)$

$B'(-5, 9)$

$C'(-9, 12)$

26 $A'(4, 6)$

$B'(-1, 5)$

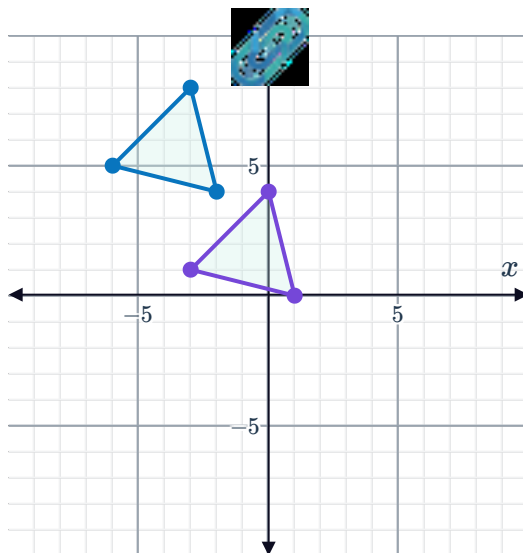
$C'(13, -12)$

27 $A'(8, -2)$

$B'(-1, 9)$

$C'(-6, 6)$

28



29

a No

b No

c No

d Yes

30

a No

b No

c No

d Yes

31

a No

b No

c Yes

d No

32

a

$A(-2, 2)$	$\rightarrow$	$A'(3, -2)$
$B(4, -3)$	$\rightarrow$	$B'(9, -7)$
$C(-4, -3)$	$\rightarrow$	$C'(1, -7)$

b

$A'(3, -2)$	$\rightarrow$	$A''(2, 3)$
$B'(9, -7)$	$\rightarrow$	$B''(7, 9)$
$C'(1, -7)$	$\rightarrow$	$C''(7, 1)$

33

a

$A(-7, -2)$	$\rightarrow$	$A'(-2, 7)$
$B(-3, -2)$	$\rightarrow$	$B'(-2, 3)$
$C(-3, -8)$	$\rightarrow$	$C'(-8, 3)$
$D(-7, -8)$	$\rightarrow$	$D'(-8, 7)$

b

$A'(-2, 7)$	$\rightarrow$	$A''(2, 7)$
$B'(-2, 3)$	$\rightarrow$	$B''(2, 3)$
$C'(-8, 3)$	$\rightarrow$	$C''(8, 3)$
$D'(-8, 7)$	$\rightarrow$	$D''(8, 7)$

34

a No

b Yes

c No

d No

35

a No

b No

c Yes

d No



36

a No

b No

c No

d Yes

37

a No

b No

c No

d Yes

38

a Yes

b No

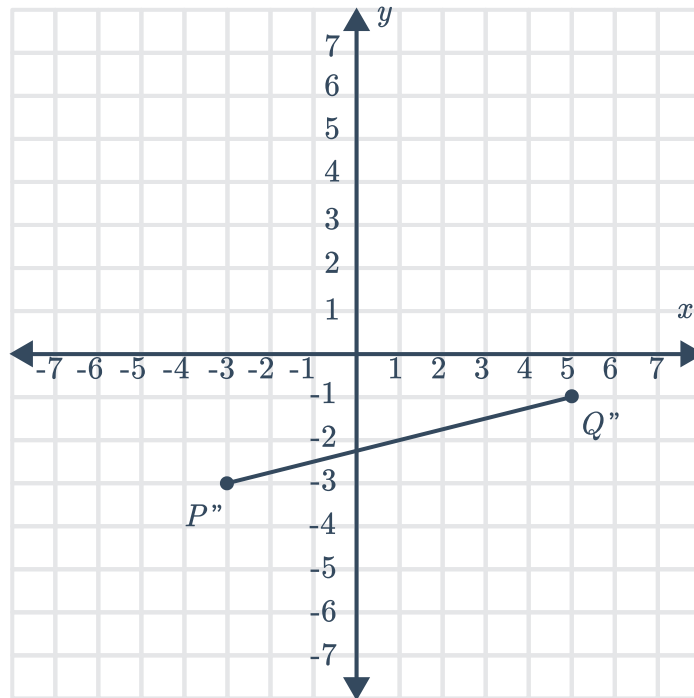
c No

d No

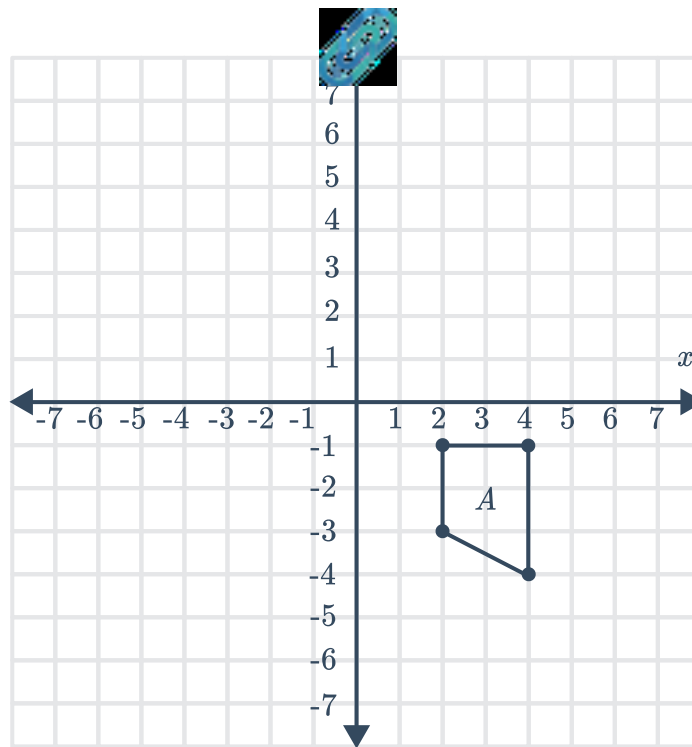
39 $(18, -18), (18, 0), (0, 0), (0, -18)$

Additional questions

40



41

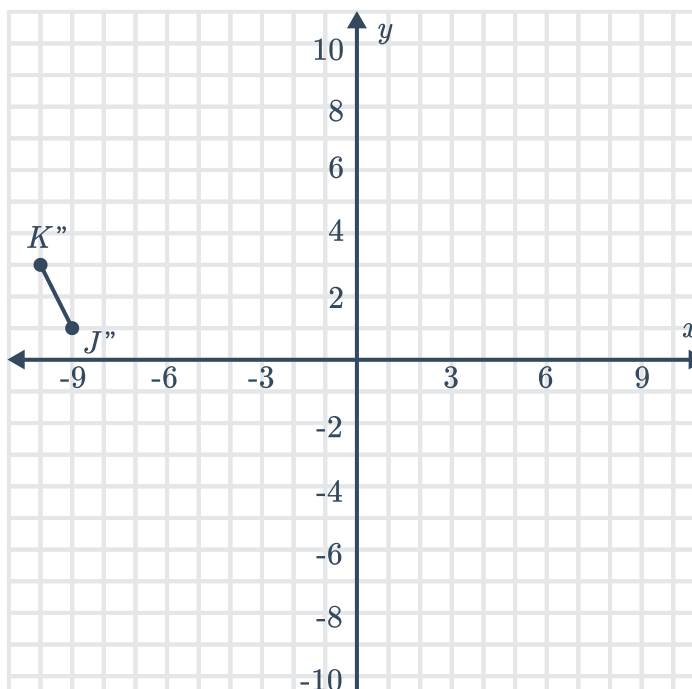


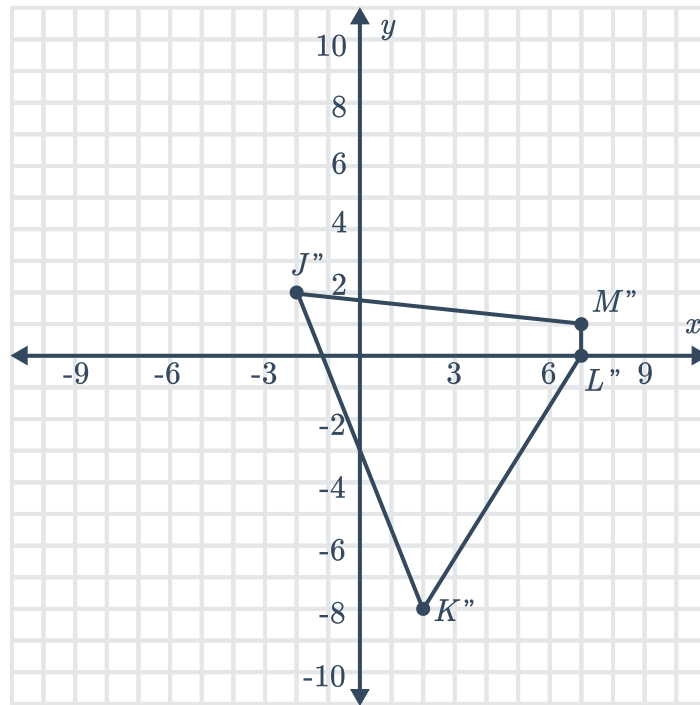
42 Reflect $\triangle ABC$ across the x -axis, then translate $\triangle A'B'C'$ by 2 units up and 6 units to the right.

43 Reflect $\triangle ABC$ across the x -axis, then rotate $\triangle A'B'C'$ by 90° counterclockwise about the origin.

44 Translate 1 unit down then reflect $\triangle ABC$ across the line $y = x$.

45





47

- a Rotate section 1 180° clockwise or counterclockwise about the center of the flag.
- b Reflect section 1 over the horizontal midline, then the vertical midline (in either order).

48 Answers may vary. Starting with the top leftmost parallelogram in the pattern, perform a series of translations down the fabric until you have six parallelograms. Then, reflect the set of six parallelograms across the line of symmetry in the pattern.